INCH-POUND
A-A-59944
24 April 2014

COMMERCIAL ITEM DESCRIPTION

CALIBRATOR-SIMULATOR, THERMOCOUPLE

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE. This Calibrator-Simulator is capable of precisely measure and emulate characteristic thermocouple voltages proportional to thermal levels. This instrument will be used for development, repair and maintenance of general electronics, medical, engine control electronics and other support equipment.

2. CLASSIFICATION. The test set covered by this Commercial Item Description (CID) should be commercially available equipment that has factory installed options and/or may be modified to the extent necessary to meet the following. The test set is Class 3, in accordance with MIL-PRF-28800.

3. SALIENT CHARACTERISTICS.

3.1 Item Description. The Simulator-Calibrator is a "Thermo-Couple Simulator" and a "Thermo-Couple" voltage reader used for the measurement, reference, comparison and calibration of thermo sensitive devices such as digital thermometers, temperature recorders and other thermal measurement instruments.

3.2 Functional Characteristics. The Simulator Calibrator shall measure and emulate thermo-couple voltage characteristic levels and responses proportional to thermal intensities. The Simulator-Calibrator shall provide a precise simulation of thermo-couple signals through the generation of accurate proportional direct current (DC) voltages from microvolt level to ±11 volts.

Comments, suggestions, or questions on this document should be addressed to: AFLCMC/GUEEA, Robins AFB GA 31098-1813. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.,mil.

AMSC N/A FSC 6625

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3.3 Thermal Range. The Simulator-Calibrator shall be capable of measuring temperature readings spanning a range equal or greater than from -270°C to 2200°C with the addition of appropriate band specific thermo-couple devices or probes.

3.4 Thermal Range Settling Time. The Simulator-Calibrator shall be capable of measuring temperature readings with a settling time of 5 s ±2 seconds at one standard atmosphere.

3.5 Thermal Reading Stability. The Simulator-Calibrator shall have a thermal reading stability deviation ≤±20ppm+3.0µV per year.

3.6 Accuracy. The Simulator-Calibrator shall be capable of measurements with a variation of ≤±25ppm ±1.5µV within a 24 hour period.

3.7 Temperature Coefficient Accuracy. The temperature frequency coefficient shall be ≤ 10ppm/°C with an error ±0.2µV/°C.

3.8 Induced Thermal Noise. The instrument induced noise shall be ≤ ±1.5µV from 0.1Hz to 10 Hz with a 95% confidence level.

3.9 Temperature Scales. The Simulator-Calibrator temperature readings shall be displayed in Celsius (°C), Rankine (°R), Fahrenheit (°F) and Kelvin (°K) scales.

3.10 Thermo-Couple Compatibility. The Simulator-Calibrator shall be capable of performing accurate measurements within the required parameters for thermo-couple types; B, C, D, E, G, J, K, N, PLII, R, S &T.

3.11 Thermo-Couple Settling Time. The Simulator-Calibrator settling time for the output voltage level in thermocouple emulation mode shall be ≤ 500 mS.

3.12 Auto-Zero Function. The Simulator-Calibrator shall have auto-zero function that sets the instrument to zero after a successful power on and self-test routine and on- demand by the user.

3.13 Output Level Conformity. The Simulator-Calibrator output level in emulation mode shall conform to NIST T/C curve 0.4µV.

3.14 Output Impedance. The Simulator-Calibrator output port impedance shall be for all thermo-coupe devices ≤ 0.75 Ω.

3.15 Output Current. The Simulator-Calibrator shall provide an output current level current ≥35 mA and be limited to ≤100mA with short circuit protection.

3.16 Display Resolution. The Simulator-Calibrator shall have a display resolution of ≥6 digits.

3.17 Display Information. The Simulator-Calibrator shall display a menu and measurement information to include scale, temperature, voltage magnitude, offset value and self-test condition and outcome.

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3.18 Data Storage. The Simulator-Calibrator shall store and display previous readings or values for later use or recall. The Simulator-Calibrator shall have a memory of 16 registers or locations that are addressable by the user.

3.19 Voltage Modes. The Simulator-Calibrator shall have a continuous auto-range mode that is enabled after a successful power on and self-test and auto-zero condition.

3.20 Voltage Reading Settling Time. The Simulator-Calibrator shall be capable of measuring thermo-couple or probe voltages with a settling time ≤1 s ±0.25 seconds.

3.21 Design and Construction. The Simulator-Calibrator shall be used in both bench top and rack mount environments. The Simulator-Calibrator is a commercial of the shelf item subject to commercial products regulatory standards. The Simulator-Calibrator shall be designed and constructed in accordance with all applicable European Union (EU) requirements with the "CE" marking affixed, International Organization for Standardization (ISO), the International Electro-technical Commission (IEC), the International Telecommunication Union (ITU), Underwriters Laboratories (UL) and the American National Standards Institute (ANSI). It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.22 System Safety. The Simulator-Calibrator shall comply with all sections of the Underwriters Laboratories (UL) standard 1449 for electrical safety. The system shall comply with MIL-STD-1472 section 5.9.11.3.3 for weight & size restrictions for single person lift & handling.

3.23 Sound Levels. The Simulator-Calibrator acoustic noise shall conform to sound level requirements on section 1 of ISO 7779 measured per section 2 of ISO 8253.

3.24 Electrostatic Discharge (ESD). The design of the Simulator-Calibrator shall preclude equipment damage due to ESD, protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking in accordance with MIL-STD-1686 section 5.2.2.3.

3.25 Electro Magnetic Interference (EMI). The Simulator-Calibrator EMI emissions shall be in accordance with the following radiated emission and susceptibility requirements of MIL-PRF-28800, MIL-STD-461: RE102 and RS103 & IEC-1000-4 thru IEC-1000-6. Emitted radiation absorption and radiation limits FCC Part 15.109 & 15.209

3.26 Service Life. The design of the Simulator-Calibrator shall be such that under normal use and operation the service repair can be performed according to MIL-PRF-28800 sections 3.14.2 & 3.14.3.

3.27 Port Damage. The output port and input ports of the Simulator-Calibrator shall withstand applied voltages ≥30 $Volts_{AC/DC}$ without damaging the internal circuitry, protection can be achieved by either internal or external fusing.

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3.28 Connectors. The Simulator-Calibrator shall have connectors complying with MIL- PRF-39012 class 1 and class 2 and category "A" connectors.

3.29 Connector Types. The Simulator-Calibrator shall have "Banana" type connectors for input and output. The Simulator-Calibrator shall have a GPIB port for control and communications protocol under IEC-60488, Standard Digital Interface for Programmable Instrumentation.

3.30 Operating Temperature Range. The Simulator-Calibrator shall meet the specifications listed herein with full capabilities within the temperature range of 0°C to 50°C after tests in MIL-PRF-28800 section 4.5.5.1.1 have been performed for class 3 equipment classification (Per table 11 on page 41).

3.31 Operating Humidity Range. The Simulator-Calibrator shall meet the specifications for non-precipitating humidity levels for both operating and not operating conditions ≤ 95% of relative humidity as specified in MIL-PRF-28800 sections 3.8.2.1 to 3.8.2.3.2 (with the applicable tolerance for density-altitude factors).

3.32 Storage Temperature Range. The Simulator-Calibrator shall meet and/or exceed manufacturers specifications for stored equipment after tests of MIL-PRF-28800 section 4.5.5.1.1 temperature, humidity & storage have been performed for instruments in the range of class 3 equipment classification (Per table 11 on page 41).

3.33 Altitude Requirements. The Simulator-Calibrator shall meet its operating performance and accuracy while operating at an altitude of 2,438 m (8,000 ft.). The equipment shall meet its performance and accuracy requirements, for operation and non-operating storage, as specified in MIL-PRF-28800 section 3.8.3 and section 4.5.5.2 respectively at this altitude.

3.34 Heat Dissipation and Ventilation. The Simulator-Calibrator ventilation shall be sufficient to allow for prevention, buildup and retention of moisture, heat dissipation either passive or active shall be suitable in order to meet MIL-PRF-28800 section 4.5.5.1.1 for class 3 equipment classification (Per table 11 on page 41).

3.35 Calibration & Maintenance Adjustments. No special tools, software, technical manuals or documentation shall be required for the alignment, adjustment, repair and calibration except those provided with the test set.

3.36 Calibration Period. The design of the Simulator-Calibrator shall provide readily accessible calibration and maintenance adjustments. The calibration adjustments shall be accessible without removal of the test set case or modules. The calibration interval shall be based on a period of one calendar year or an operating time of 2,080 hours, whichever occurs first. At the end of this interval 85% of the units shall remain within calibration tolerances in accordance with MIL-PRF-28800 section 3.12.

3.37 Reliability. The design of the Simulator-Calibrator shall be such that under normal use and operation the mean time before failure (MTBF) exceeds 2,080 hours of operation with a statistical certainty equal or greater than 95% in accordance with MIL-PRF-28800 section 3.13. The manufacturer shall provide a summary of maintenance or warranty records to establish the validity of the statistical assertion in accordance with MIL-PRF-28800 section 3.13. The manufacturer shall be able to reasonably establish the claim that the offered equipment is capable of the required level of reliability by meeting and/or exceeding MIL-PRF-28800 sections 3.13 and section 4.5.8.1.

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3.38 Built in Test Equipment and Diagnostics. The Simulator-Calibrator shall be equipped with built in test and diagnostic capability that at “Power On or “Cold Start” performs a system test evaluating all functional characteristic of the instrument in accordance with MIL-PRF-28800 section 6.5.2.6.

3.39 Maintainability. The Simulator-Calibrator shall meet the human engineering factors of ergonomics for operational & maintainability criteria, providing access to controls, internal components and adjustments in accordance with MIL-STD-1472 paragraphs 5.9 through 5.9.18 and MIL-PRF-28800 section 6.6.2.4.

3.40 Preventive Maintenance. Preventive maintenance shall not require more than 15 minutes per 30-day period. Preventive maintenance shall not require breaking of the equipment seams where calibration seals would normally be placed in accordance with MIL-PRF-28800 section 3.14.2.

3.41 Corrective Maintenance. The Simulator-Calibrator shall be designed to meet reparability criterion in accordance with section 3.14.1 of MIL-PRF-28800.

3.42 Electrical Power Sources & Connections. The Simulator-Calibrator equipment shall operate from nominal commercial, military and shipboard power sources with a nominal voltage of 120 V_{RMS} Single Phase AC (±10%) at line frequencies of 50Hz, 60Hz, in accordance with MIL-PRF-28800 paragraph 3.1.5.2 and 3.1.5.2a.

3.43 Electrical Surge. The Simulator-Calibrator shall be able to provide surge protection that meets section 38 of the UL 1449 standard for equipment and personnel protection.

3.44 Noise Suppression. The Simulator-Calibrator shall be able to operate with electrical noises and harmonic content present on the alternating current power source with frequencies in the 30Hz to 50 kHz range and levels varying from line level of 15 Volts measured at the power input in accordance with UL 1283, IEC/EN 60939 and under the guidance of IEEE-519 standard.

3.45 Dimensions. The Simulator-Calibrator shall comply with width and depth of EIA/ECA-310-E standard equipment rack configuration. The height shall be compatible with no more than 2 U’s. One unit or "U" is defined by EIA/ECA-310-E to be 1.75 inches (44.45 mm).

3.46 Weight. The total weight of the Simulator-Calibrator, excluding accessories and manuals, shall comply with MIL-STD-1472 section 5.9.11.3.3 for weight & size restrictions for single person lift & handling and shall not exceed a one person lift of 9 Kg (20 pound).

3.47 Rack Mount. The Simulator-Calibrator shall be capable of being rack-mounted in a standard EIA/ECA-310-E equipment rack and if required include a rack-mount conversion kit that meets the requirements of MIL-PRF-28800 section 3.6.4.14.1.

3.48 Markings. All warning decals and instrument category rating must conform to commercial safety standards and be visible by user in accordance with MIL-STD-130. Markings, Information/Caution must be Lusterless Black, and Markings, Warning/Danger shall be Lusterless Red.

4. REGULATORY REQUIREMENTS.

4.1 Recycled Recovered Materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the

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operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and the product conforms to the producer's engineering drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Source of Documents.

7.1.1 Military Specifications, Standards and Handbooks referenced herein may be obtained at https://assist.dla.,mil or available from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg. 4, Section D, Philadelphia, PA 19111-5094.

7.1.2 FAR. FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/ .

7.1.3 EIA/ECA documents can be obtained online at http://www.eciaonline.org/default.aspx .

7.1.4 IEEE copies can be obtained online at www.**ieee**.org .

7.1.5 Underwriters Laboratories (UL) copies may be obtained online at http://ulstandardsinfonet.ul.com or from: COMM 2000, 1414 Brook Drive, Downers Grove, IL 60515-5000.

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7.1.6 IEC documents may be obtained at http://www.eciaonline.org/default.aspx or from Electronic Components Industry Association, 111 Alderman Drive, Suite 400, Alpharetta, GA 30005.

7.1.7 ISO documents can be obtained online at http://www.iso.org or from the International Organization for Standardization, ISO Central Secretariat, 1, ch de la Voie-Creuse, CP 56-CH-1211 Geneva 20, Switzerland.

7.1.8 ANSI/ASME documents may be obtained online at http://www.asme.org/ or from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

7.1.9 ITU copies can be obtained at http://www.itu.int/en/Pages/default.aspx.

7.2 Key Words.

Maintenance
Repair
Temperature
Voltage

Custodians:
Air Force – 84

Reviewers:
Air Force – 99

Preparing Activity:
Air Force – 84

Agent:
Air Force – 99

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